Victoria Xu

5th F Sir Ernst Chain Building, Imperial College London – London, UK
Email: vx18@ic.ac.uk • Website: https://v1ckyxu.github.io/V1ckyXu/

Overview

Structural biology postdoctoral research associate at Imperial College London investigating virus DNA packaging mechanism using cryoEM.

Employment

Postdoctoral Research Associate **Imperial College London**
London, UK *May 2026 - Present*
Sarah L. Rouse lab

- Received funding from UKRI Impact Acceleration Account to work on a translational project based on work produced during the PhD.

Research Assistant **Academia Sinica**
Taipei, Taiwan *Jul 2020 - Nov 2020*
Cheng-Hsun Ho lab

Education

PhD in Life Sciences **Imperial College London**
Oct 2022 - May 2026

Supervisors: Sarah L. Rouse, Doryen Bubeck and Matthew Wake.
Examiners: Christian Speck (Imperial), Jonathan Grimes (Oxford).
Funding: UKRI-BBSRC and AstraZeneca.

- Structural biology research carried out in pharmaceutical and academic labs as well as national cryo-EM facility (eBIC).

MRes in Molecular and Cellular Biosciences **Imperial College London**
Distinction *Oct 2021 - Oct 2022*

- Two 5-month-long research projects: 1. Studying mitochondrial inner-membrane fusion mediated by OPA1 using cryoEM SPA and cgMD simulations in Sarah L. Rouse's group. 2. Studying cellular defence against complement proteins by CD59 using confocal microscopy in Doryen Bubeck's group.

BSc in Biochemistry **Imperial College London**
2.1 Honors *Oct 2018 - Oct 2021*

Publications, Grants and Patents

- Xu, V., McInnes, A., Wake, M., Acebrón-García-de-Eulate, M., Barritt, J. D., Bubeck,D., Rouse, S. L. (2026). Structural basis for Rep-mediated adeno-associated virus packaging. *Cell Reports*, 45(3), 117044. https://doi.org/10.1016/j.celrep.2026.117044
- Named Investigator on UKRI Impact Acceleration Account (IAA) grant proposal.
- Lead inventor on UK patent application
- Reviewed papers for Nature Communications and American Chemical Society.

Conferences

Talks:

- Flatiron institute Structural and Molecular Biophysics (SMBp) seminar speaker. New York, NY, USA. 2025 (https://indico.flatironinstitute.org/event/4067/)
- Biophysical Society (BPS) annual meeting platform speaker. Los Angeles, CA, USA. 2025 (https://www.cell.com/biophysj/abstract/S0006-3495(24)00992-5)
- Imperial Centre for Structural Biology open day speaker. London, UK. 2025

Session chair:

- BPS annual meeting protein-nucleic acid platform I. Los Angeles, CA, USA. 2025

Posters:

- The Collaborative Computational Project for Electron cryo-microscopy (CCP-EM) Spring symposium. Nottingham, UK. 2023-2026.
- American Society for Gene and Cell Therapy (ASGCT). New Orleans, LA, USA. 2025 (https://www.sciencedirect.com/science/article/pii/S1525001625003648)
- British Crystallographic Association winter meeting. Oxford, UK. 2023
- LMB-EPFL graduate student symposium. Cambridge, UK 2026

Awards

- Eurofins Foundation Postgraduate Prize for MRes (Top student).
- Imperial Department of Life Sciences Travel Award to visit Flatiron institute.
- CCPEM bursary to attend 2026 CCPEM spring symposium.
- The Life Sciences European Studies Fund to attend 2026 ESGCT conference.

Technical Experience

- **Wet lab:** Microscope operation (Glacios and Krios, EPU software), grid-freezing (Vitrobot Mark IV, Leica GP2, Chameleon high-speed grid vitrification robot), cloning, protein expression and purification (E.coli, mammalian cells, yeast), x-ray crystallography, mass photometry, confocal microscopy, ELISA, and AAV virus production.
- **Dry lab:** Coarse-grained molecular dynamics simulation (CHARMM-GUI, GROMACS); SPA image processing (cryoSPARC and RELION), model building (ISOLDE, Coot, phoenix and ModelAngelo), deposition (PDB, EMDB and EMPIAR), Linux command line.
- **Teaching:** Supervised Structural Biology MRes students over 10-month wet-lab project, Chemistry MRes student over 10-month wet-lab project and Biochemistry undergraduates over 3-month wet- or dry-lab projects.
- **Reviewing:** Co-reviewed manuscripts submitted to Molecular Pharmaceutics (published doi: 10.1021/acs.molpharmaceut.5c01279.), Nature Communications and Molecular Therapy Advances.